R122 D2 | full18 locked outcome

R122/model | same BASE f342aedf161db09a0883 | update 14400

Split	Top1 %	mean_cost	SBS	Oracle	vsSBS %	vsOracle %	Regret %
TRAIN	68.52757	11.40747	11.40070	11.34507	-0.08533	0.50092	0.49452
VAL	68.44815	11.43526	11.42798	11.37169	-0.08157	0.50744	0.50298
TEST	68.62346	11.39241	11.38530	11.32993	-0.08252	0.50208	0.49595

No feasible VAL checkpoint: constraints NOT satisfied
Actual full TRAIN constraints satisfied: False; selected VAL constraints satisfied: False
TRAIN budgets update the controller; separate VAL thresholds select the checkpoint only. No dual gradient uses VAL.
Error/regret ALL is task-equal. Mean raw cost is descriptive; vsSBS/Oracle ratios differ from per-instance actual regret.
TEST was development-exposed. No current TEST/LIB checkpoint/rule selection. No solver quotas.
NSS uses historical70114-trained weights, not a new90k retrain. Histories and missing references are disclosed.
Observed D2 joint TEST evidence vs R84A: False

What D2 optimizes, constrains and selects

R122/model | same BASE f342aedf161db09a0883 | update 14400

Component	Frozen meaning
Primal objective	Native winner CE/log(2) plus multiplier-weighted normalized full-pool regret proxy and fixed36 CE proxy
TRAIN hard feedback	Equal-task actual hard-action regret and unconditional fixed36 missed frequency; one commit after each successful logical update
TRAIN budgets	Saved selected R84A TRAIN only; fixed historical36 targets, not dynamic zero groups or quotas
VAL selection	Separate saved R84A VAL thresholds; no VAL dual updates. Feasible: native Top1, regret, earliest. Infeasible fallback: normalized violations first
Inference	One unchanged original R45A selector; maximin; stable global-ID score ties; no current labels/costs in forward
Evidence boundary	Proxy/controller heuristic, no deterministic nonconvex feasibility guarantee; TEST development-exposed; no per-task assembly

The actual objective/controller parameters are recorded in protocol.json without modification.
Selected feasibility/fallback is reported even when constraints fail; completing PDFs is not a scientific success claim.

TRAIN | ALL and18 tasks

R122/model | same BASE f342aedf161db09a0883 | update 14400

Task	Top1 %	Cost	SBS	Oracle	vsSBS %	vsOracle %	Regret %
ALL	68.52757	11.40747	11.40070	11.34507	-0.08533	0.50092	0.49452
TSP	43.22346	8.00542	8.02396	7.97022	-0.23099	0.44177	0.44299
CVRP	15.60864	18.35250	18.15199	17.94682	1.10461	2.26046	2.14490
ATSP	69.74074	1.64267	1.67868	1.63394	-2.14510	0.53434	0.53542
OVRP	39.05432	7.44624	7.45353	7.39768	-0.09776	0.65653	0.67006
VRPB	80.32963	9.35693	9.35726	9.31934	-0.00353	0.40338	0.41083
VRPL	35.24321	12.39625	12.40088	12.34450	-0.03732	0.41922	0.42693
VRPTW	80.27037	18.81337	18.81370	18.75795	-0.00178	0.29542	0.30910
OVRPTW	83.82099	10.11836	10.11864	10.09322	-0.00272	0.24912	0.24493
OVRPB	88.36296	6.36319	6.36323	6.34666	-0.00069	0.26042	0.26861
OVRPL	39.81975	7.44637	7.45322	7.39848	-0.09186	0.64729	0.66062
VRPBL	79.52593	9.40285	9.40312	9.36320	-0.00280	0.42353	0.42482
VRPBTW	79.52346	19.73228	19.73355	19.64271	-0.00642	0.45600	0.44052
VRPLTW	80.42099	18.76926	18.76996	18.71437	-0.00374	0.29331	0.30772
OVRPBL	88.55062	6.35887	6.35886	6.34251	0.00025	0.25794	0.26535
OVRPBTW	83.28395	10.62387	10.62445	10.58623	-0.00540	0.35562	0.33093
OVRPLTW	83.76420	10.11100	10.11127	10.08577	-0.00263	0.25017	0.24723
VRPBLTW	79.60741	19.76962	19.77081	19.68036	-0.00604	0.45353	0.43570
OVRPBLTW	83.34568	10.62532	10.62554	10.58737	-0.00207	0.35853	0.33467

native int64 ind verified against original FP64 costs; never FP32 argmin
SBS is fitted on full81000 TRAIN only; Oracle is the per-instance legal-pool minimum.
vsSBS=100*(mean_cost/SBS_cost-1); vsOracle=100*(mean_cost/Oracle_cost-1).
Actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]. No per-task model assembly.

VAL | ALL and18 tasks

R122/model | same BASE f342aedf161db09a0883 | update 14400

Task	Top1 %	Cost	SBS	Oracle	vsSBS %	vsOracle %	Regret %
ALL	68.44815	11.43526	11.42798	11.37169	-0.08157	0.50744	0.50298
TSP	44.24444	8.02493	8.04535	7.99053	-0.25390	0.43043	0.42688
CVRP	15.02222	18.47674	18.26938	18.06409	1.13505	2.28440	2.18131
ATSP	69.66667	1.64135	1.67690	1.63247	-2.12044	0.54374	0.54082
OVRP	39.51111	7.46322	7.46958	7.41578	-0.08508	0.63982	0.65502
VRPB	80.86667	9.35518	9.35488	9.31875	0.00326	0.39095	0.40010
VRPL	35.48889	12.43427	12.43796	12.38344	-0.02963	0.41047	0.41828
VRPTW	79.95556	18.81892	18.81943	18.76614	-0.00269	0.28128	0.29669
OVRPTW	83.80000	10.14329	10.14373	10.11686	-0.00431	0.26128	0.25799
OVRPB	88.51111	6.35725	6.35725	6.34108	0.00000	0.25510	0.26501
OVRPL	40.57778	7.43942	7.44571	7.39042	-0.08443	0.66311	0.67755
VRPBL	78.53333	9.36983	9.37152	9.32837	-0.01810	0.44445	0.44788
VRPBTW	78.42222	19.94907	19.95043	19.85108	-0.00683	0.49359	0.47435
VRPLTW	78.75556	18.76622	18.76587	18.70784	0.00184	0.31203	0.32935
OVRPBL	89.20000	6.37009	6.37009	6.35282	0.00000	0.27184	0.28358
OVRPBTW	83.55556	10.66055	10.66110	10.62324	-0.00519	0.35112	0.32490
OVRPLTW	83.33333	10.15808	10.15742	10.12982	0.00647	0.27900	0.28686
VRPBLTW	79.35556	19.79644	19.79704	19.70642	-0.00307	0.45679	0.44401
OVRPBLTW	83.26667	10.60983	10.60996	10.57130	-0.00115	0.36449	0.34300

native int64 ind verified against original FP64 costs; never FP32 argmin
SBS is fitted on full81000 TRAIN only; Oracle is the per-instance legal-pool minimum.
vsSBS=100*(mean_cost/SBS_cost-1); vsOracle=100*(mean_cost/Oracle_cost-1).
Actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]. No per-task model assembly.

TEST | ALL and18 tasks

R122/model | same BASE f342aedf161db09a0883 | update 14400

Task	Top1 %	Cost	SBS	Oracle	vsSBS %	vsOracle %	Regret %
ALL	68.62346	11.39241	11.38530	11.32993	-0.08252	0.50208	0.49595
TSP	44.06667	7.97310	7.98962	7.93759	-0.20679	0.44731	0.44560
CVRP	15.55556	18.32723	18.12498	17.91851	1.11583	2.28099	2.16825
ATSP	69.13333	1.63827	1.67421	1.62944	-2.14648	0.54186	0.54342
OVRP	39.57778	7.44846	7.45575	7.40049	-0.09775	0.64825	0.65848
VRPB	80.48889	9.35083	9.35181	9.31530	-0.01043	0.38148	0.38546
VRPL	35.53333	12.35187	12.35614	12.30096	-0.03457	0.41386	0.42114
VRPTW	80.95556	18.69024	18.69055	18.63741	-0.00164	0.28346	0.29502
OVRPTW	83.93333	10.09290	10.09306	10.06814	-0.00158	0.24601	0.23659
OVRPB	88.08889	6.37104	6.37083	6.35223	0.00333	0.29615	0.30397
OVRPL	40.00000	7.46622	7.47254	7.41803	-0.08461	0.64958	0.66389
VRPBL	79.42222	9.39605	9.39638	9.35597	-0.00347	0.42845	0.43161
VRPBTW	78.75556	19.66681	19.66573	19.57767	0.00552	0.45533	0.45394
VRPLTW	79.26667	18.80393	18.80314	18.74514	0.00424	0.31363	0.32642
OVRPBL	89.26667	6.37579	6.37574	6.35931	0.00077	0.25915	0.27044
OVRPBTW	84.84444	10.65936	10.65959	10.62475	-0.00219	0.32575	0.29752
OVRPLTW	82.93333	10.08512	10.08611	10.05900	-0.00981	0.25969	0.25883
VRPBLTW	79.82222	19.70393	19.70707	19.61328	-0.01592	0.46217	0.44499
OVRPBLTW	83.57778	10.66216	10.66215	10.62557	0.00010	0.34432	0.32161

native int64 ind verified against original FP64 costs; never FP32 argmin
SBS is fitted on full81000 TRAIN only; Oracle is the per-instance legal-pool minimum.
vsSBS=100*(mean_cost/SBS_cost-1); vsOracle=100*(mean_cost/Oracle_cost-1).
Actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]. No per-task model assembly.

TRAIN constraints vs separate VAL thresholds

R122/model | same BASE f342aedf161db09a0883 | update 14400

Split	Constraint	Observed %	Reference %	Satisfied	Role
TRAIN	hard_macro_actual_regret	0.49452	0.39521	False	TRAIN controller budget
TRAIN	F36 missed_frequency	5.70302	5.69150	False	TRAIN controller budget
VAL	hard_macro_actual_regret	0.50298	0.40349	False	VAL checkpoint selection only; no dual gradie
VAL	F36 missed_frequency	5.70247	5.69630	False	VAL checkpoint selection only; no dual gradie
TEST	hard_macro_actual_regret	0.49595	0.40349	False	descriptive TEST vs frozen VAL reference; NOT
TEST	F36 missed_frequency	5.66296	5.69630	True	descriptive TEST vs frozen VAL reference; NOT

Selected VAL feasible=False; fallback=True
TEST threshold comparisons are descriptive, NOT a selection gate or dual-update target.
Proxy-Lagrangian organization is a heuristic here, not a deterministic nonconvex feasibility guarantee.

Natural versus fixed36 native-winner error budgets

R122/model | same BASE f342aedf161db09a0883 | update 14400

Split	Group	Native winners	Missed	Population %	Error frequency %	Regret contribution %
TRAIN	FIXED36	83279	83150	5.71187	5.70302	0.08919
TRAIN	OUTSIDE36	1374721	375718	94.28813	25.76941	0.40533
TRAIN	FULL18	1458000	458868	100.00000	31.47243	0.49452
VAL	FIXED36	4625	4619	5.70988	5.70247	0.09048
VAL	OUTSIDE36	76375	20938	94.29012	25.84938	0.41249
VAL	FULL18	81000	25557	100.00000	31.55185	0.50298
TEST	FIXED36	4598	4587	5.67654	5.66296	0.09211
TEST	OUTSIDE36	76402	20828	94.32346	25.71358	0.40384
TEST	FULL18	81000	25415	100.00000	31.37654	0.49595

F36 missed frequency is unconditional over natural instances, NOT conditional recall.
FIXED36 + OUTSIDE36 reconstruct FULL18; task-equal macro includes tasks with zero fixed population.

One current BASE versus historical NSS | four exact datasets

R122/model | same BASE f342aedf161db09a0883 | update 14400

Dataset	n	NSS Top1 %	R122 Top1 %	NSS mean_cost	R122 mean_cost	Status
TSP TEST	4500	45.88889	44.06667	7.97040	7.97310	complete
CVRP TEST	4500	45.97778	15.55556	18.00480	18.32723	complete
TSPLIB	49	20.40816	26.53061	8.21791	8.20577	complete
CVRPLIB	100	70.00000	4.00000	67.10111	69.45764	complete

NSS: historical70114 TRAIN/old VAL selection; stored predictions on the exact new4500 TEST and original LIBs.
No NSS retraining or new inference. Native labels are primary; old3895 TEST metrics are never reused.
All four current cells use the same checkpoint SHA f342aedf161db09a0883264d0eb2a003b63de93d4da4174a9f7fde332240ad6b

Native LIB | same current checkpoint

R122/model | same BASE f342aedf161db09a0883 | update 14400

Dataset	Top1 %	Cost	TRAIN SBS	Oracle	vsSBS %	vsOracle %	Regret %
TSPLIB	26.53061	8.20577	8.24044	8.14217	-0.42071	0.78112	0.84724
CVRPLIB	4.00000	69.45764	67.24536	66.94509	3.28986	3.75315	3.32601

Standard raw input/geometry pipeline; full selector factory and frozen TRAIN buffers; no LIB refit or selection.
Legacy epoch is derived in memory from the unique update; best.pt and model parameters/buffers are never rewritten.

Same90k historical comparisons | not newly matched training controls

R122/model | same BASE f342aedf161db09a0883 | update 14400

Reference / split	Corrected	Harmed	Net TP	dTop1 pp	dCost	dRegret pp
R122/model vs R84A VAL	2393	4529	-2136	-2.63704	0.01865	0.09948
R122/model vs R84A TEST	2417	4533	-2116	-2.61235	0.01856	0.10111
R122/model vs R94B VAL	2576	4990	-2414	-2.98025	0.01898	0.10390
R122/model vs R94B TEST	2565	4958	-2393	-2.95432	0.02017	0.11684

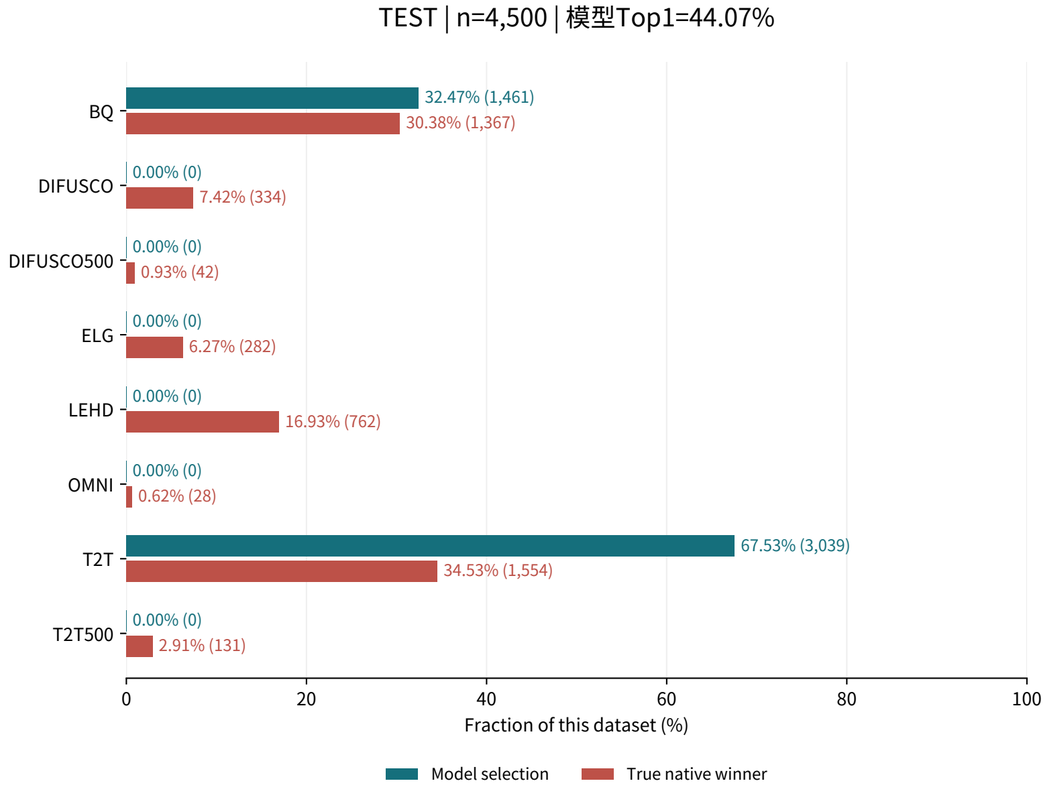
R84A model/architecture matched, but budget/loss/R-Drop not matched; R94B model_matched=false (pretrained node fusion).
Positive dTop1 is better; positive dRegret/dCost is worse. Selection counts are not the claim.
Reference status: R84A=complete; R94B=complete

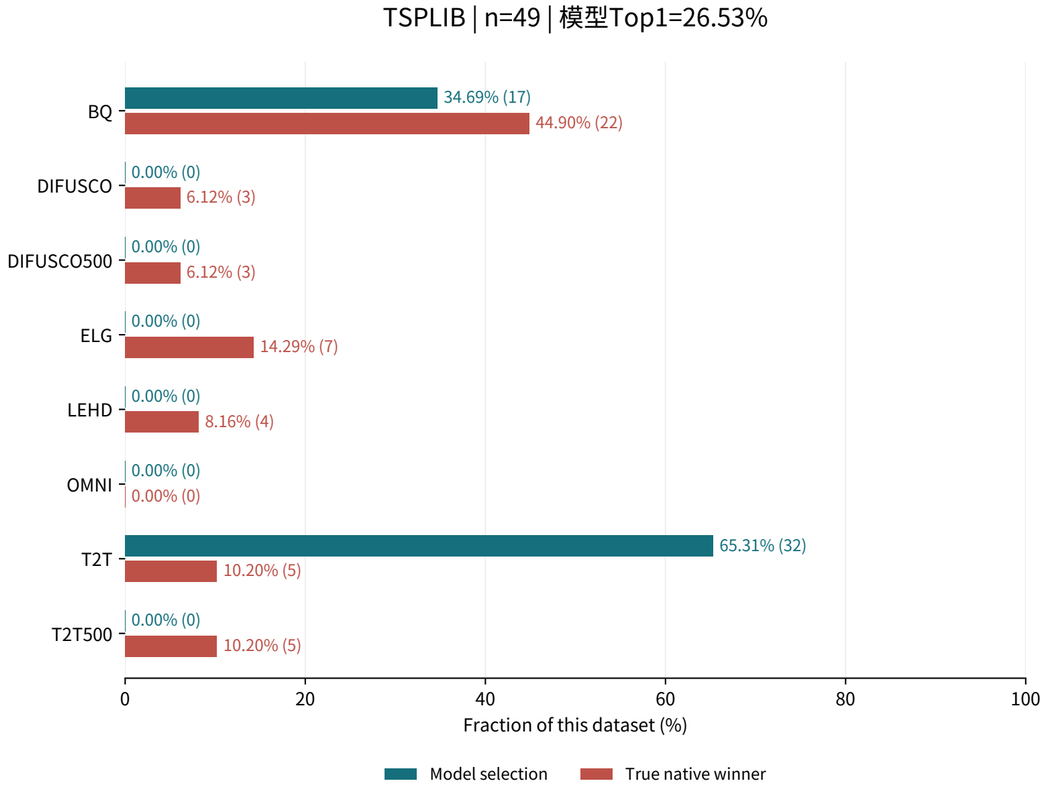
R122/model | TSP | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB=original native TSPLIB | native int64 ind verified against original FP64 costs; never FP32 argmin





Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

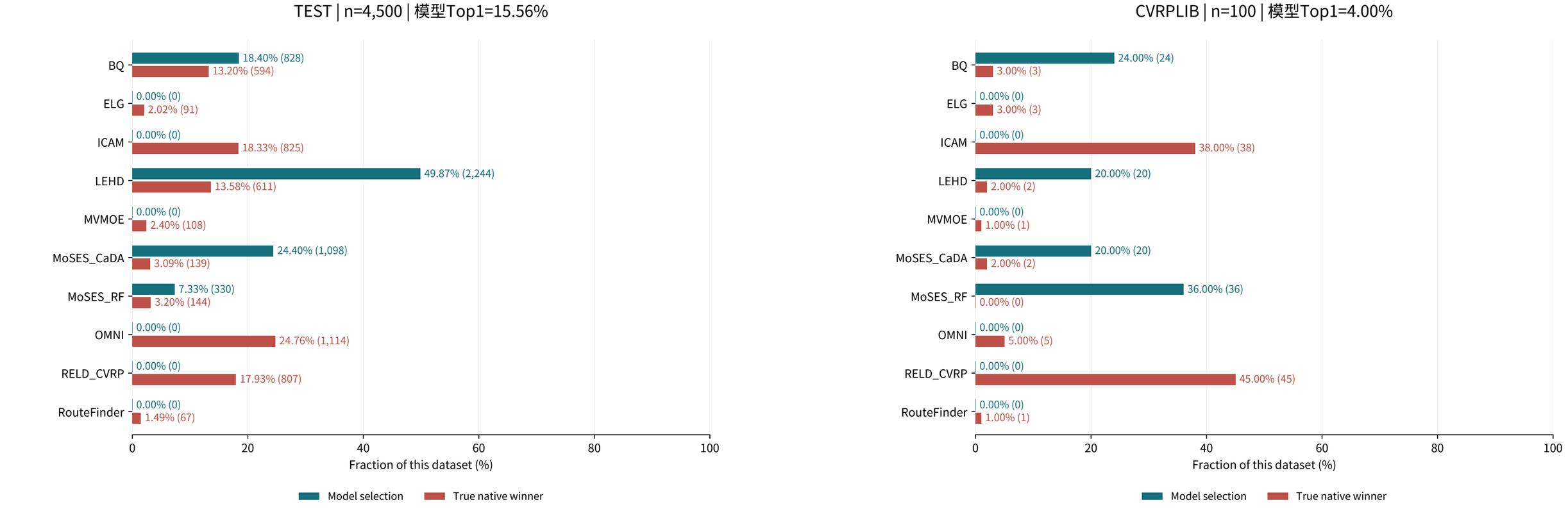
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | CVRP | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB=original native CVRPLIB | native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

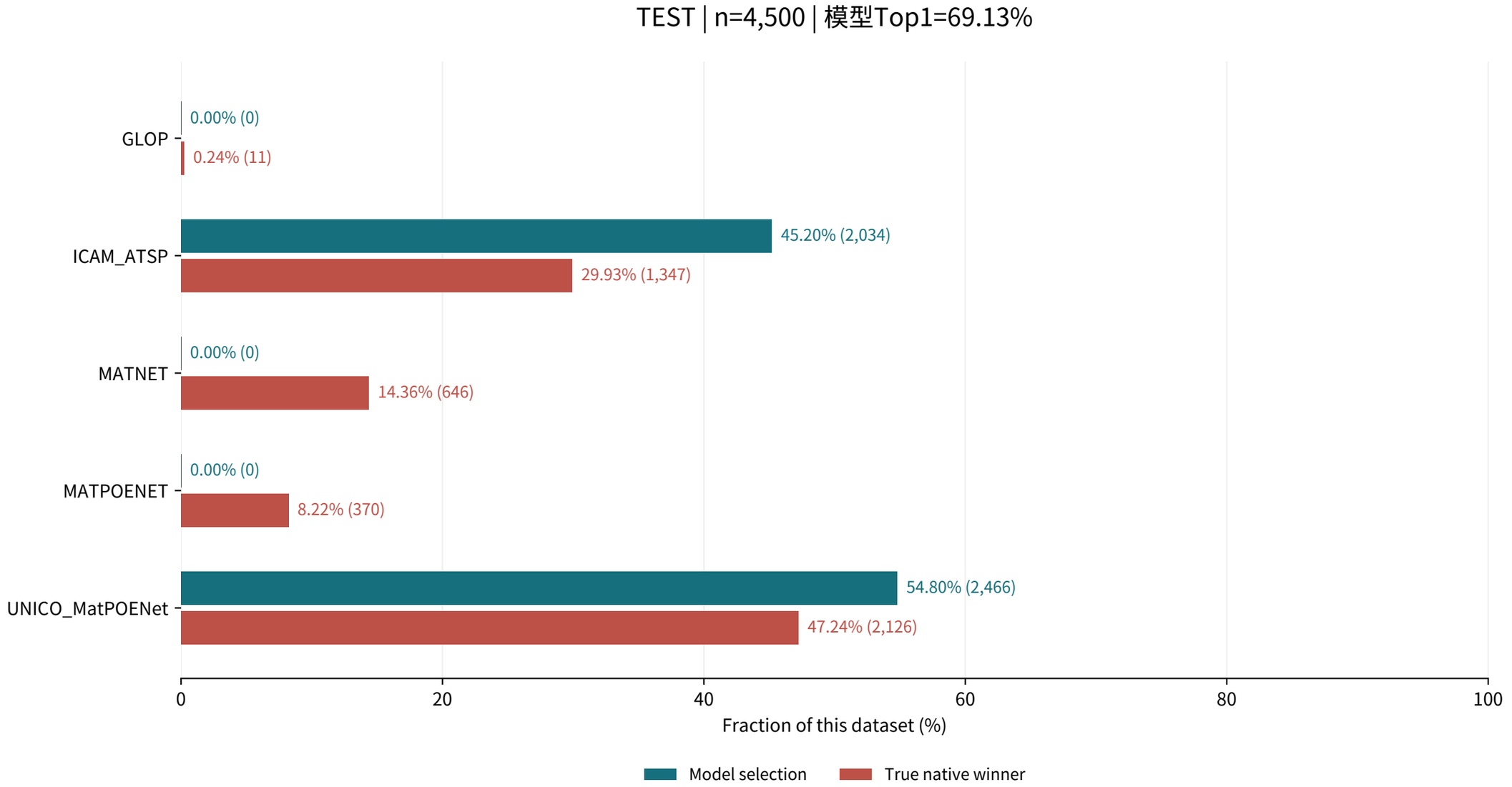
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | ATSP | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

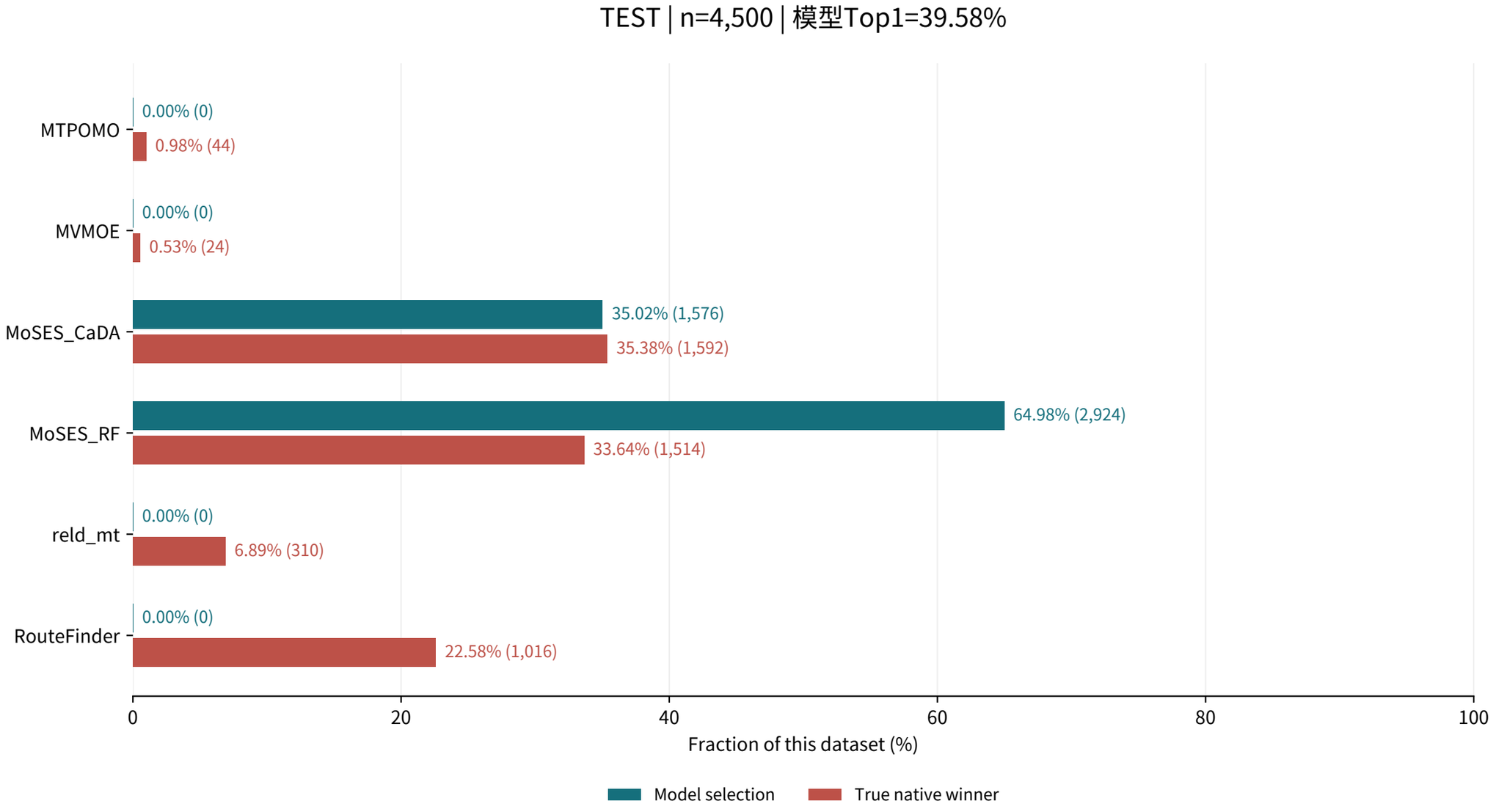
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRP | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

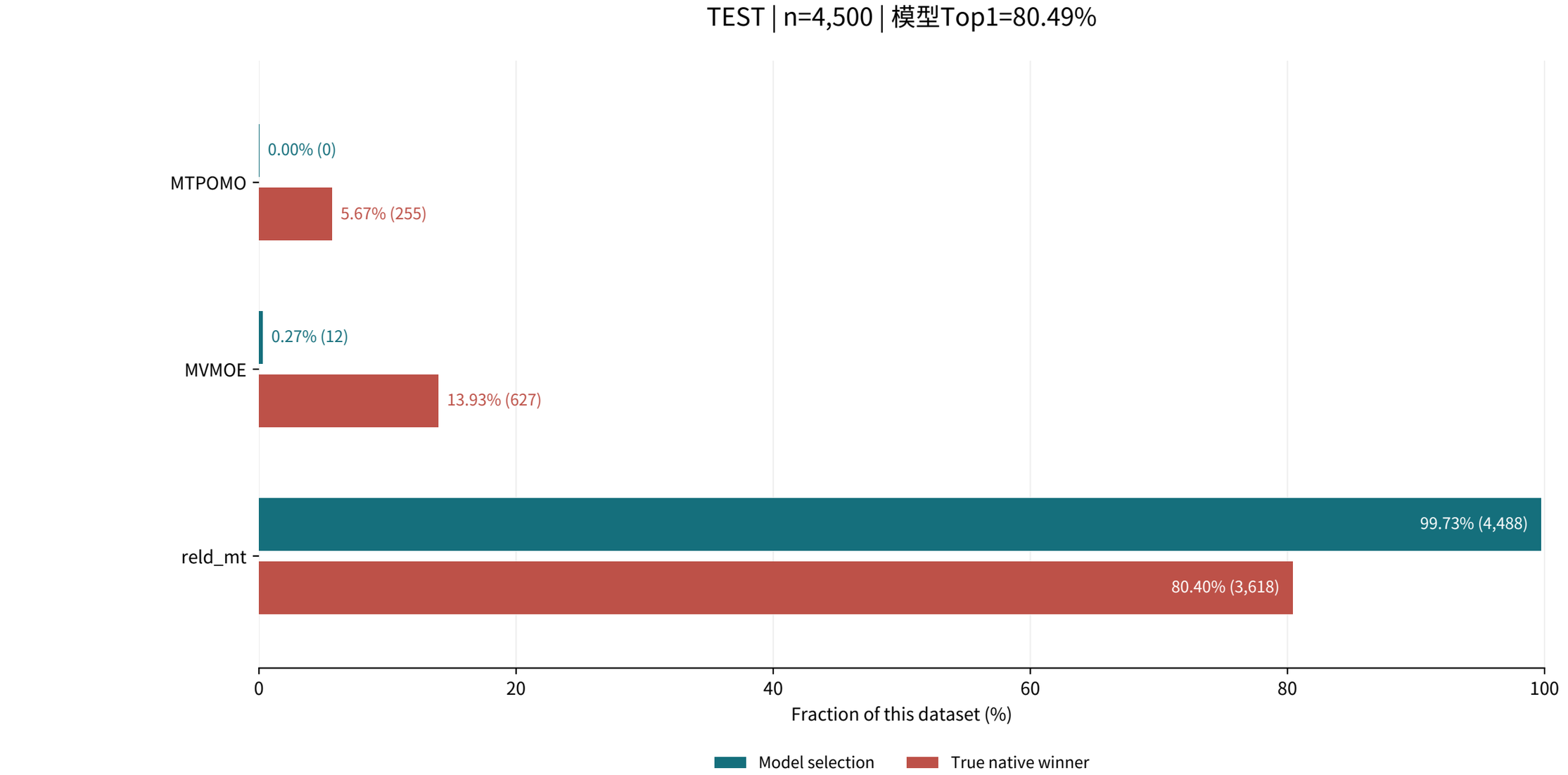
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | VRPB | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

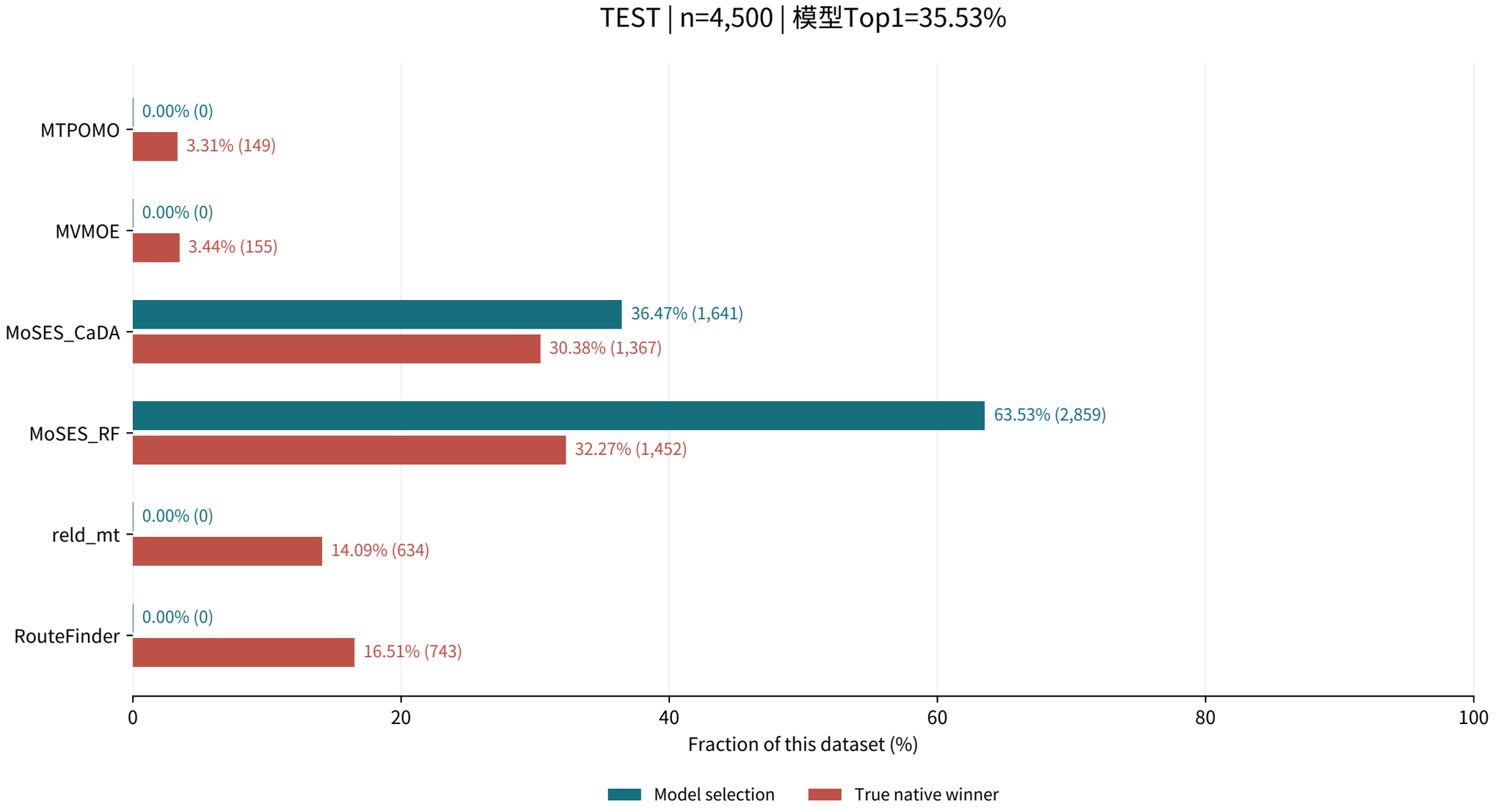
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | VRPL | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

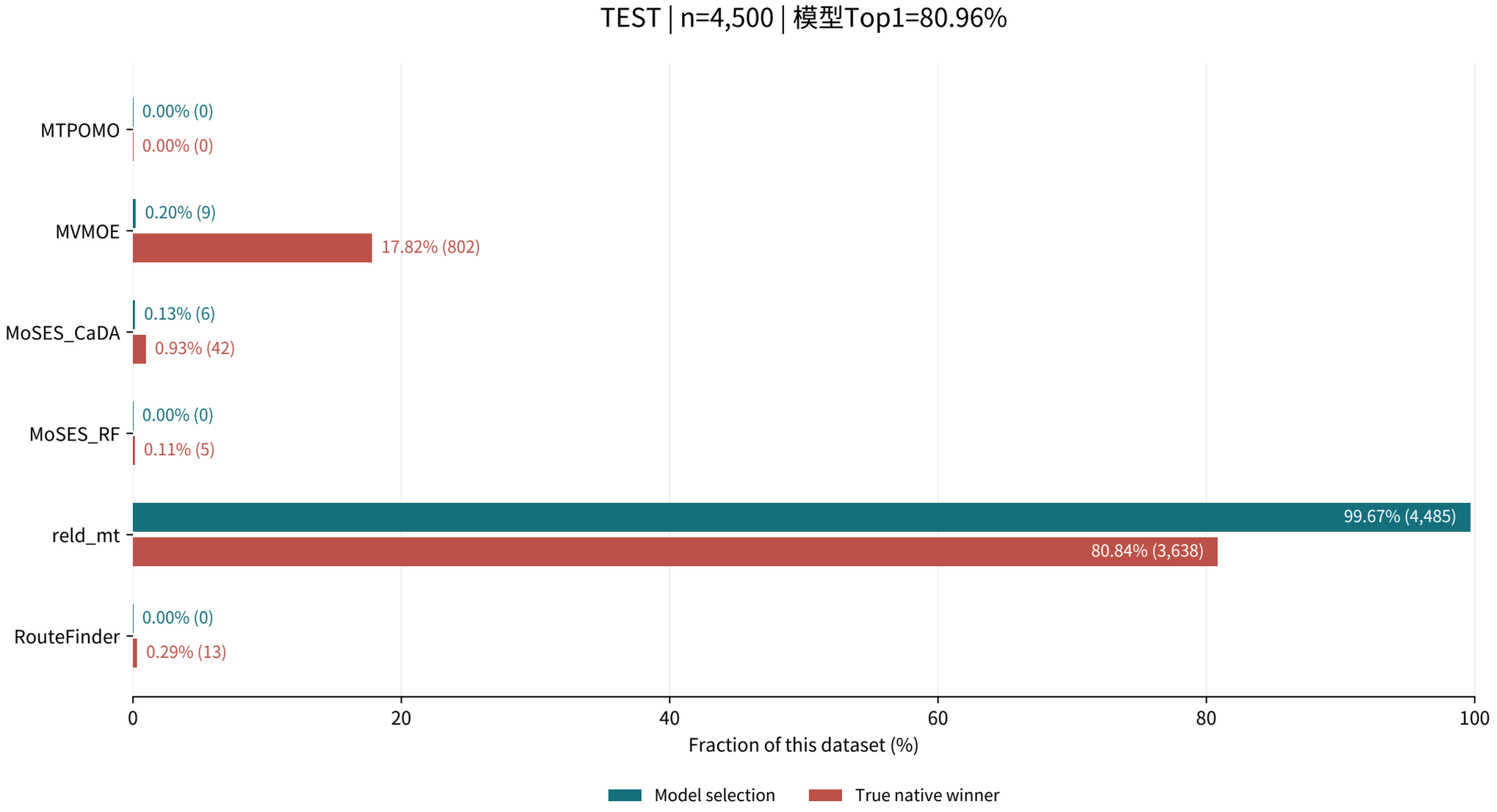
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | VRPTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

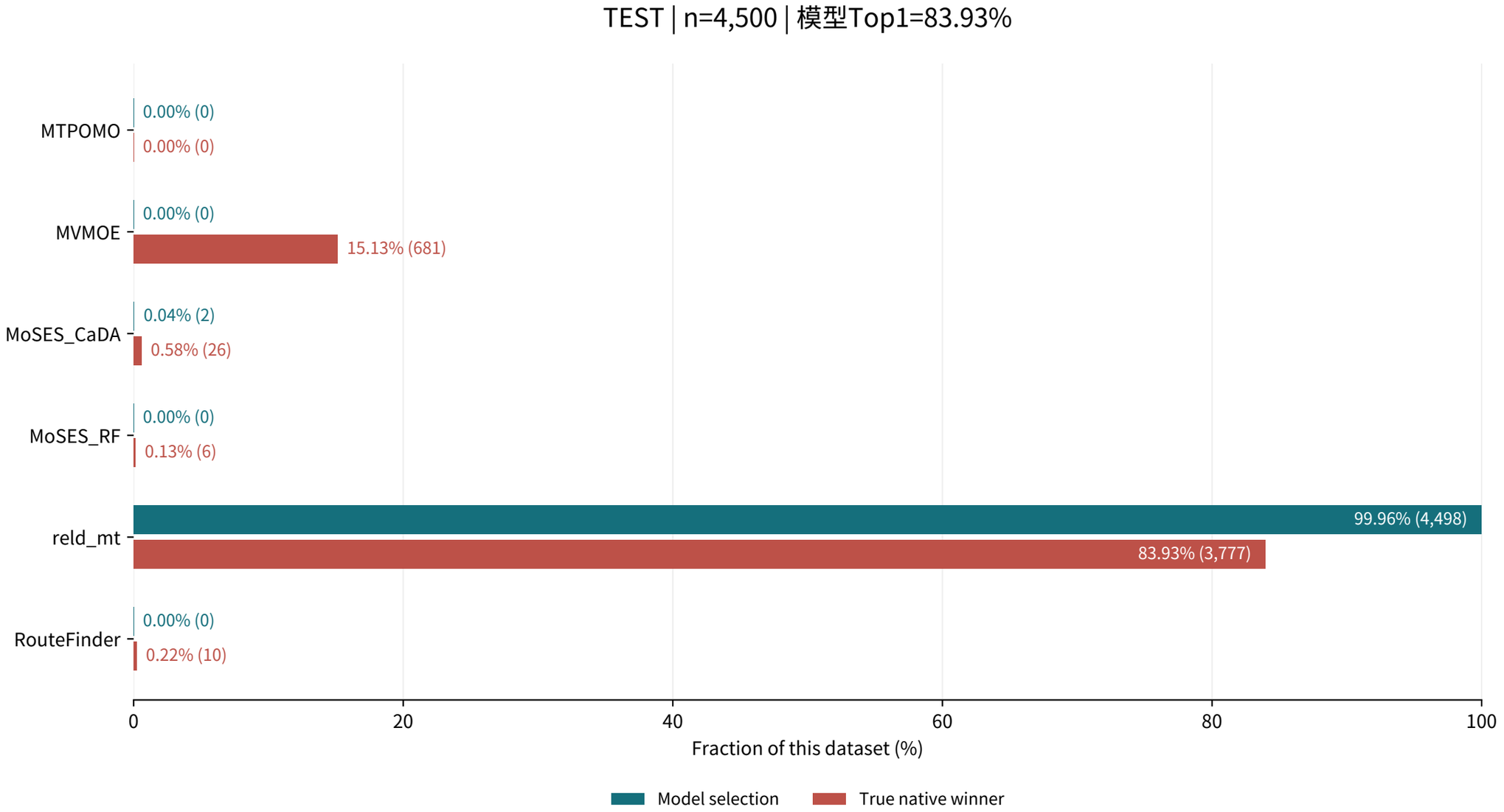
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRPTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

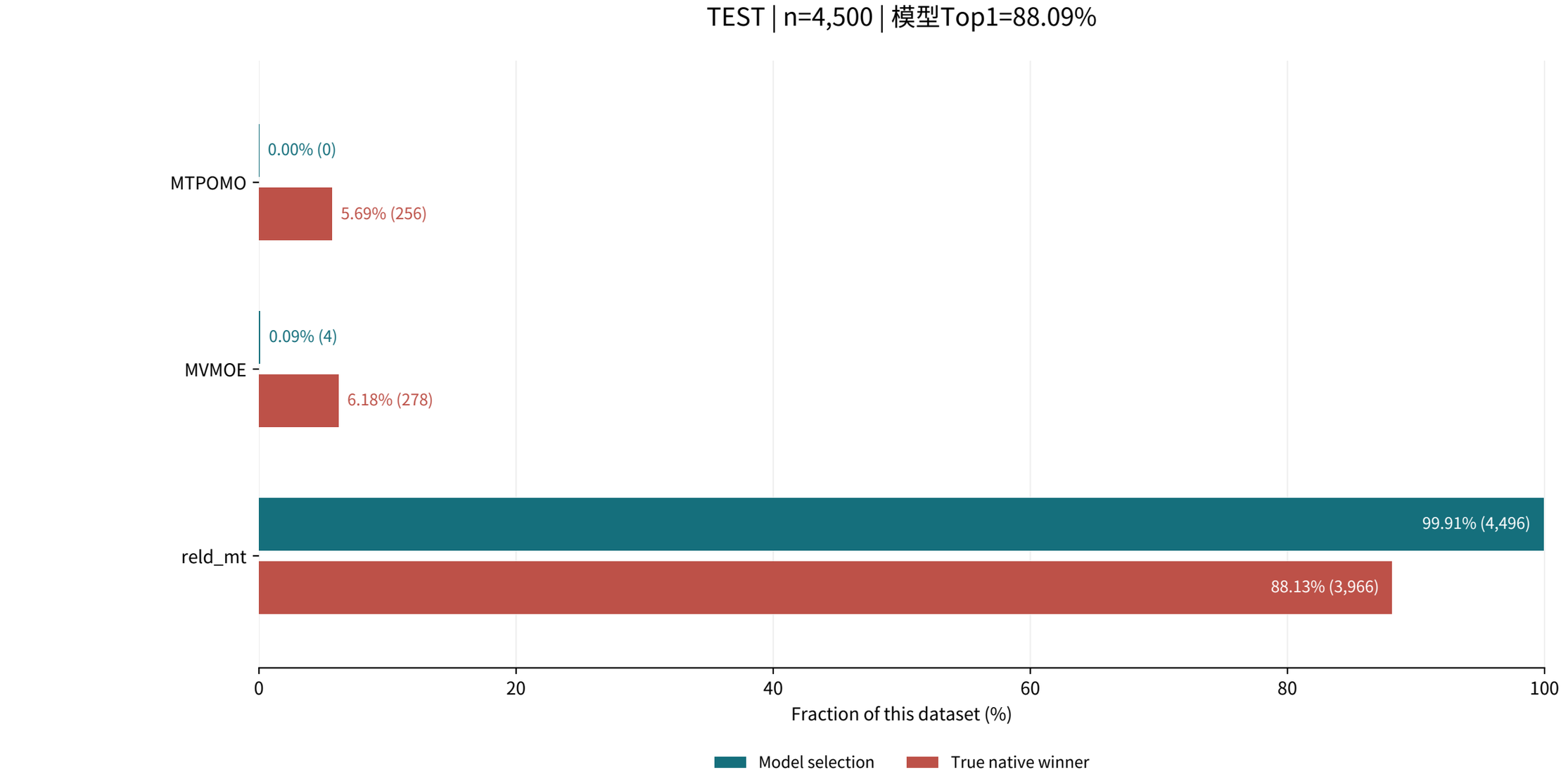
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRPB | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

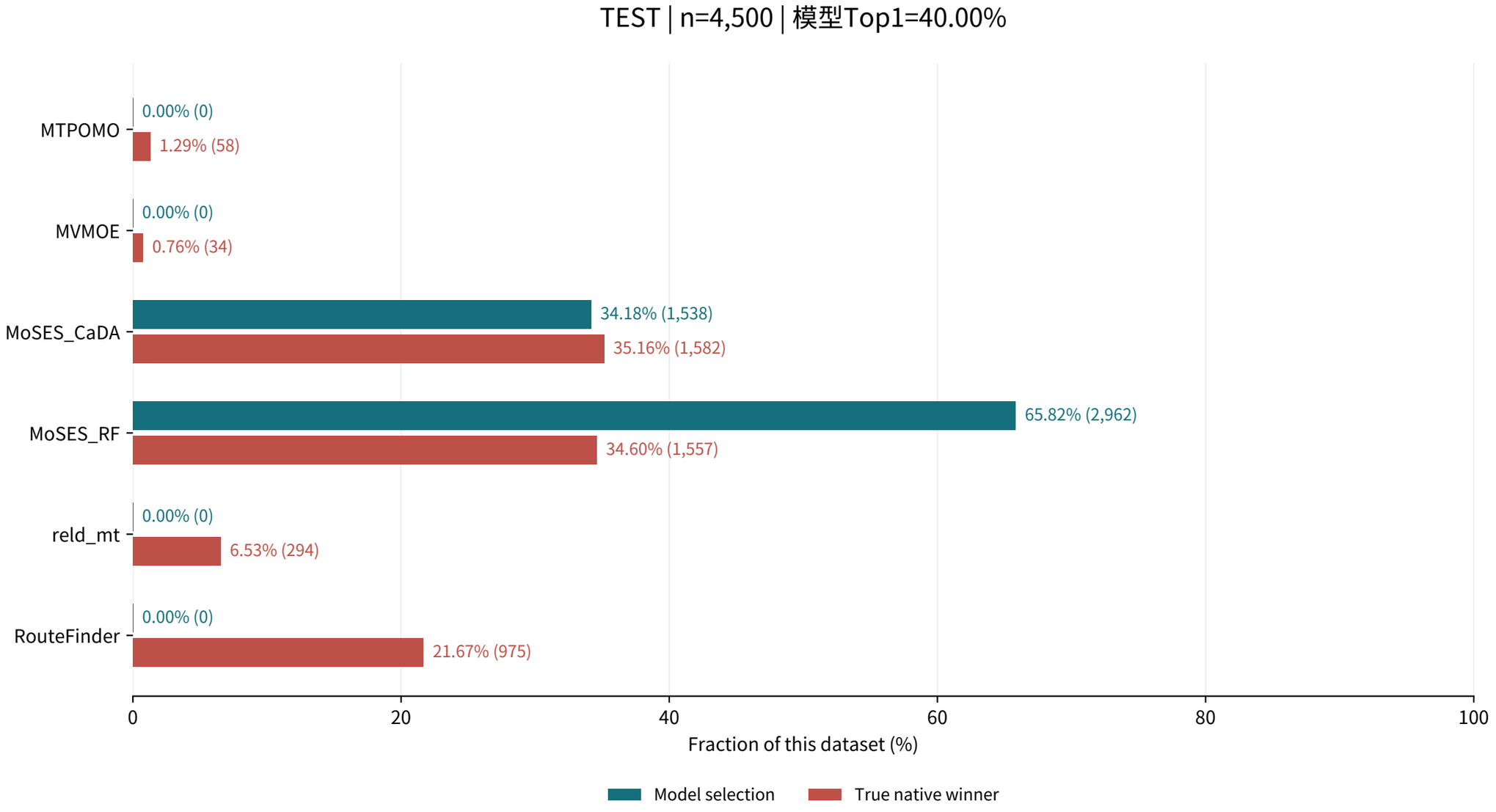
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRPL | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

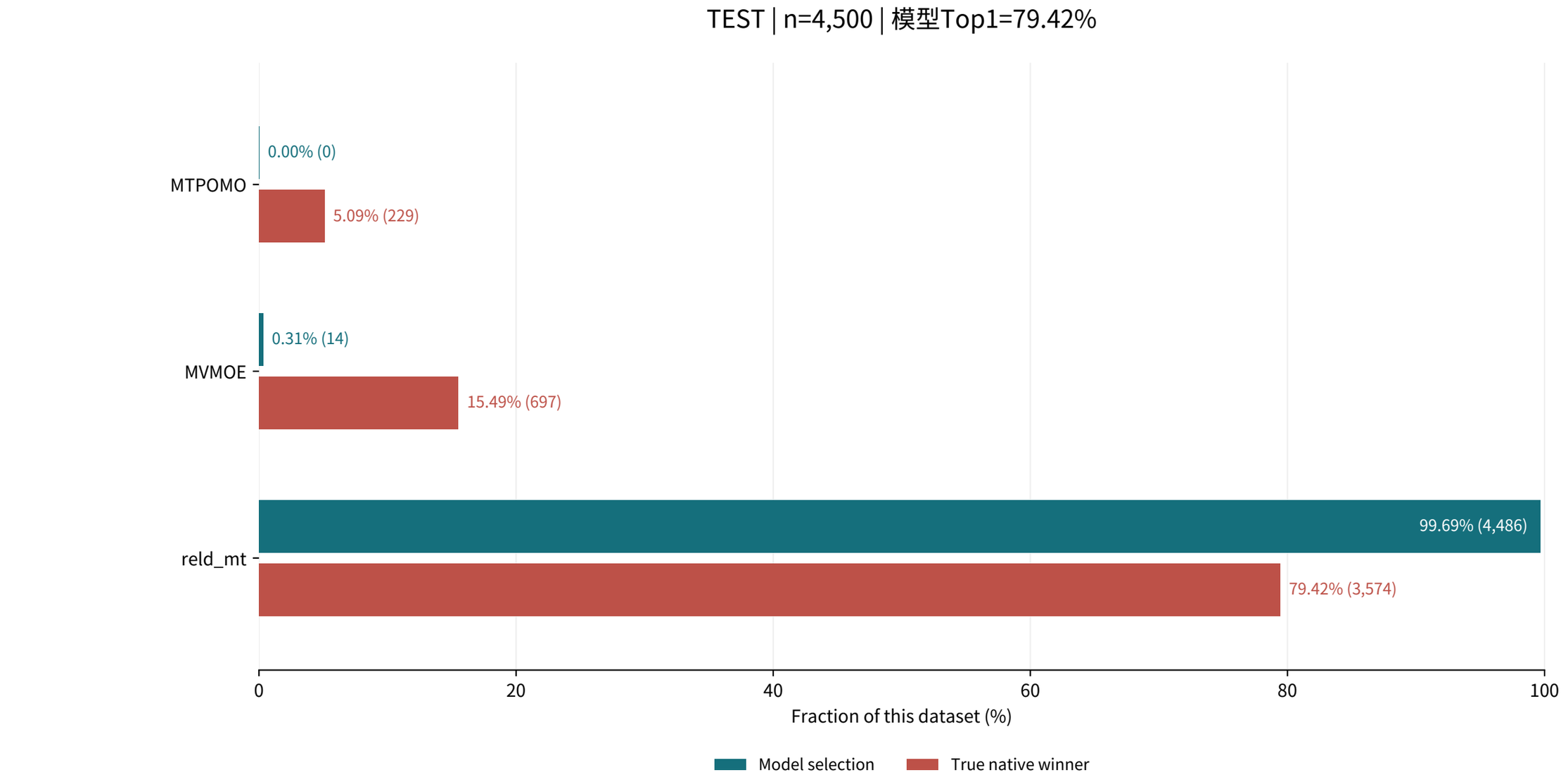
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | VRPBL | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

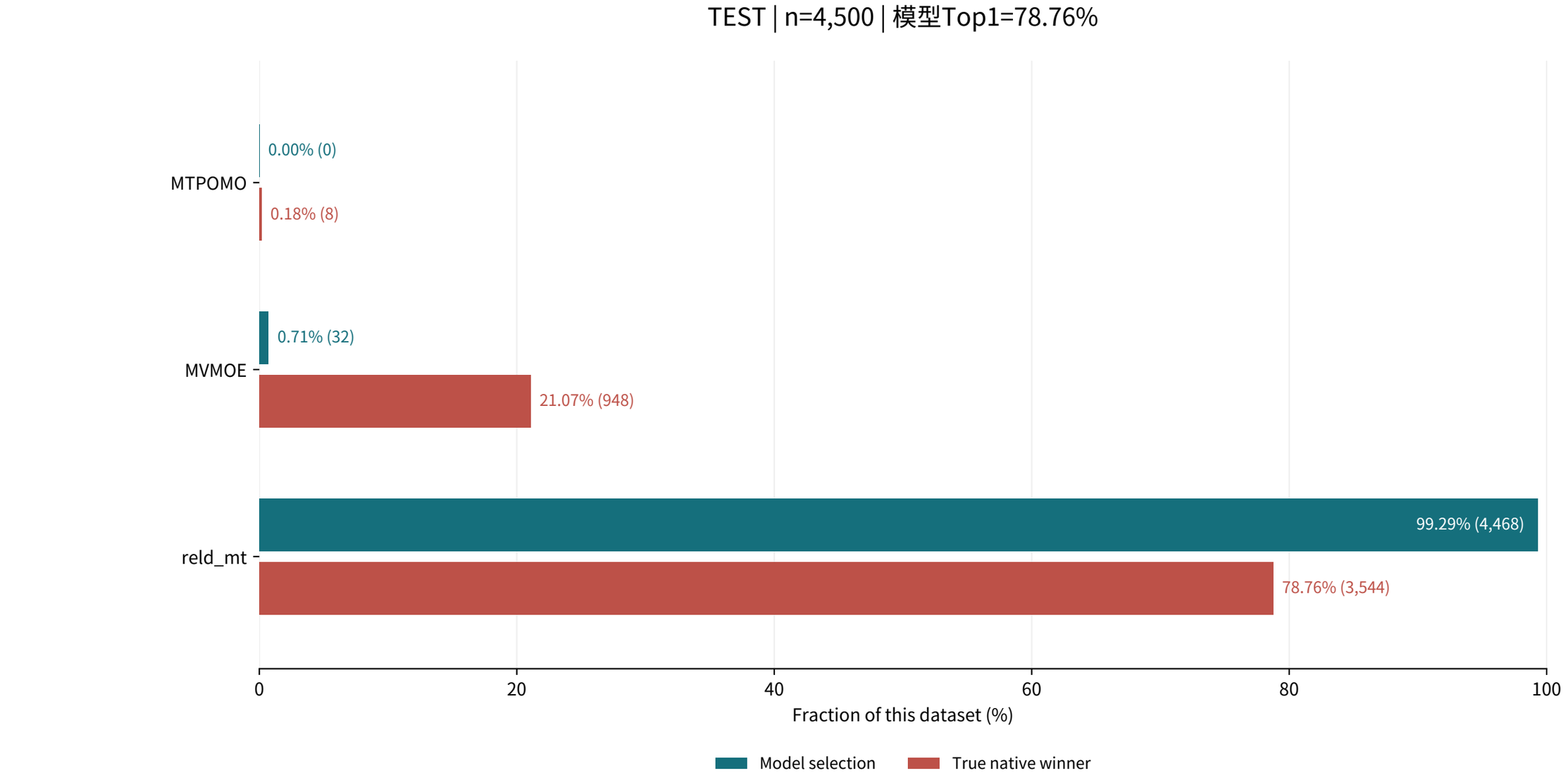
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | VRPBTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

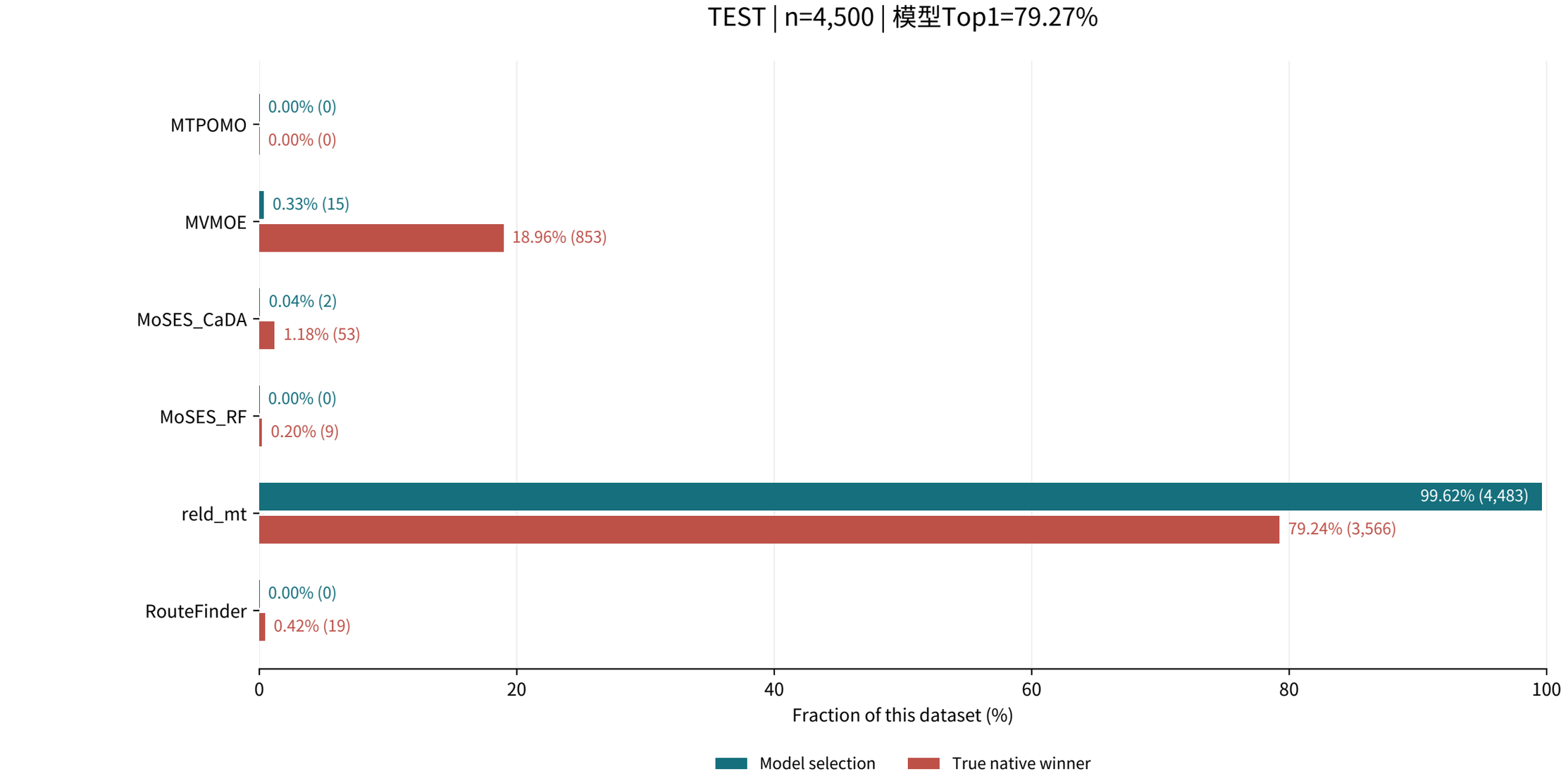
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | VRPLTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

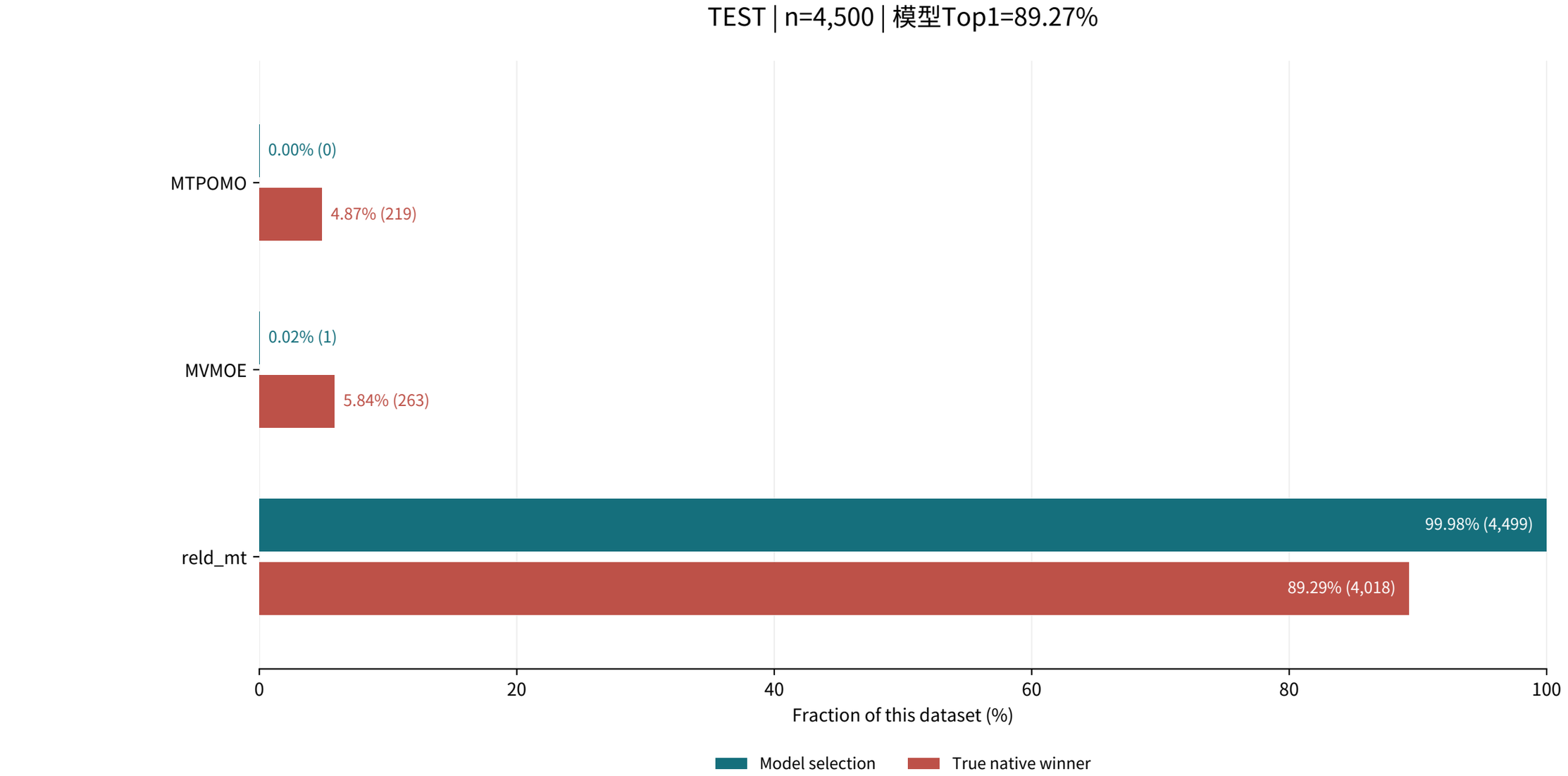
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRPBL | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

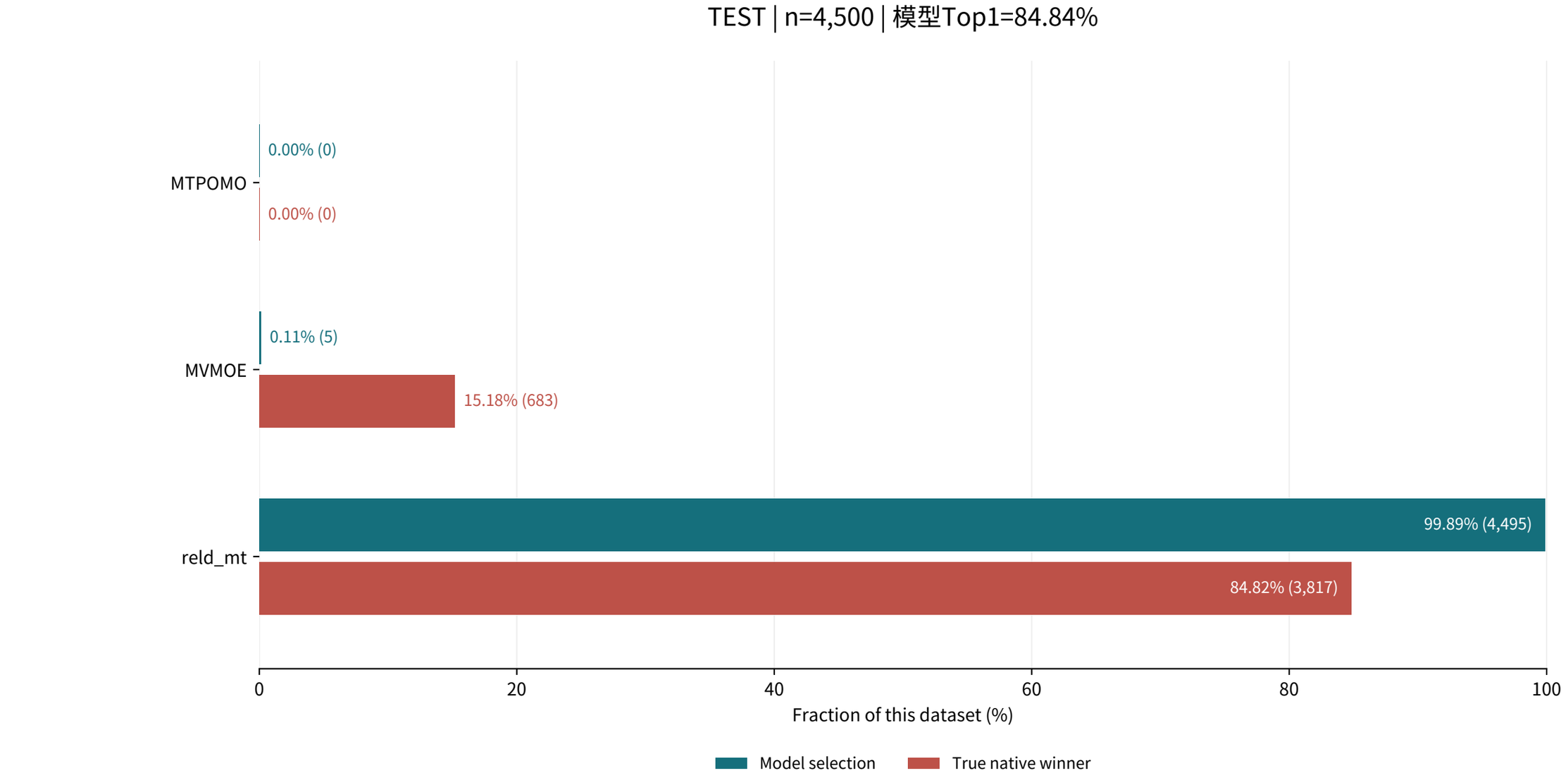
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRPBTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

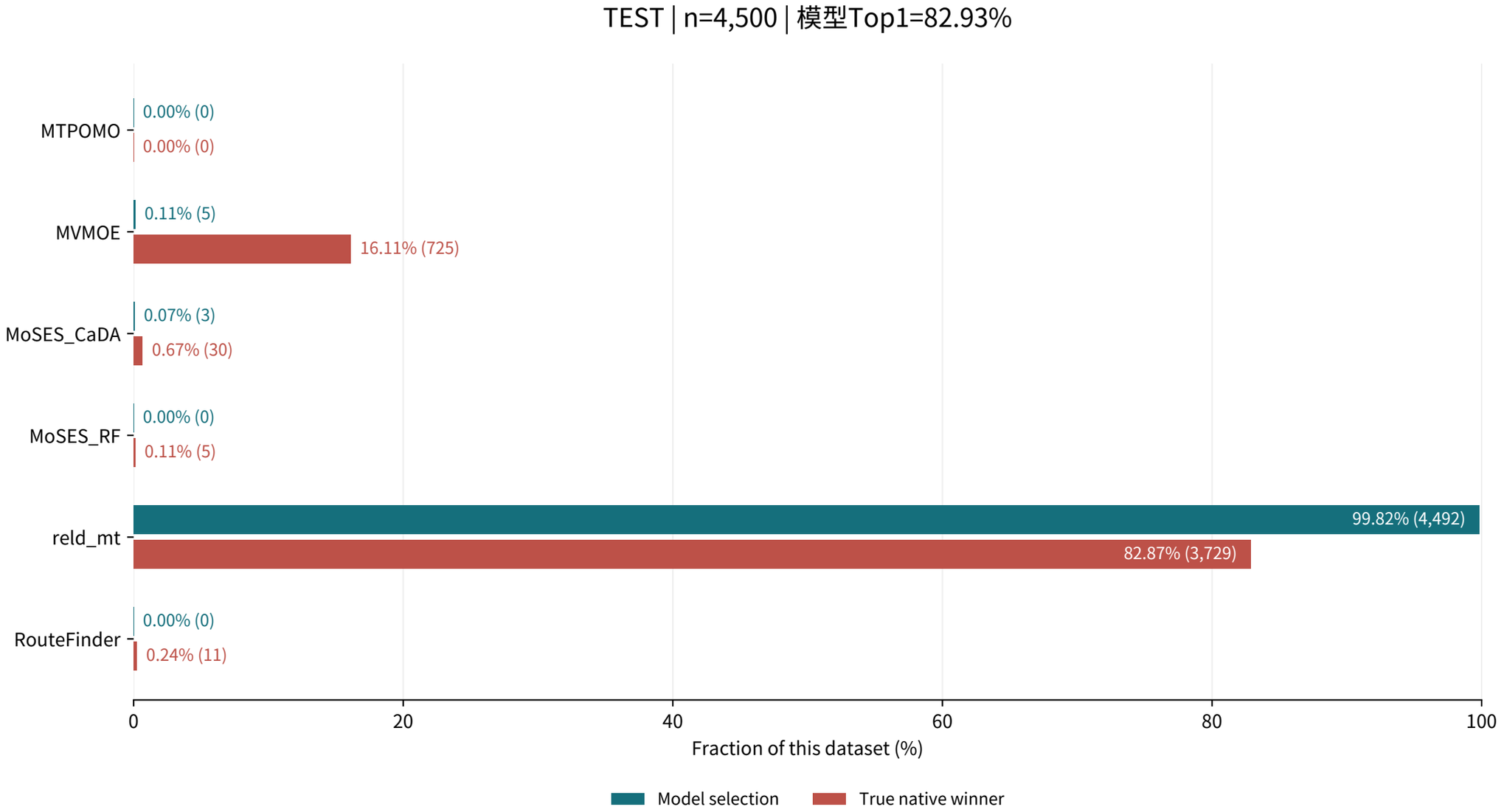
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRPLTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

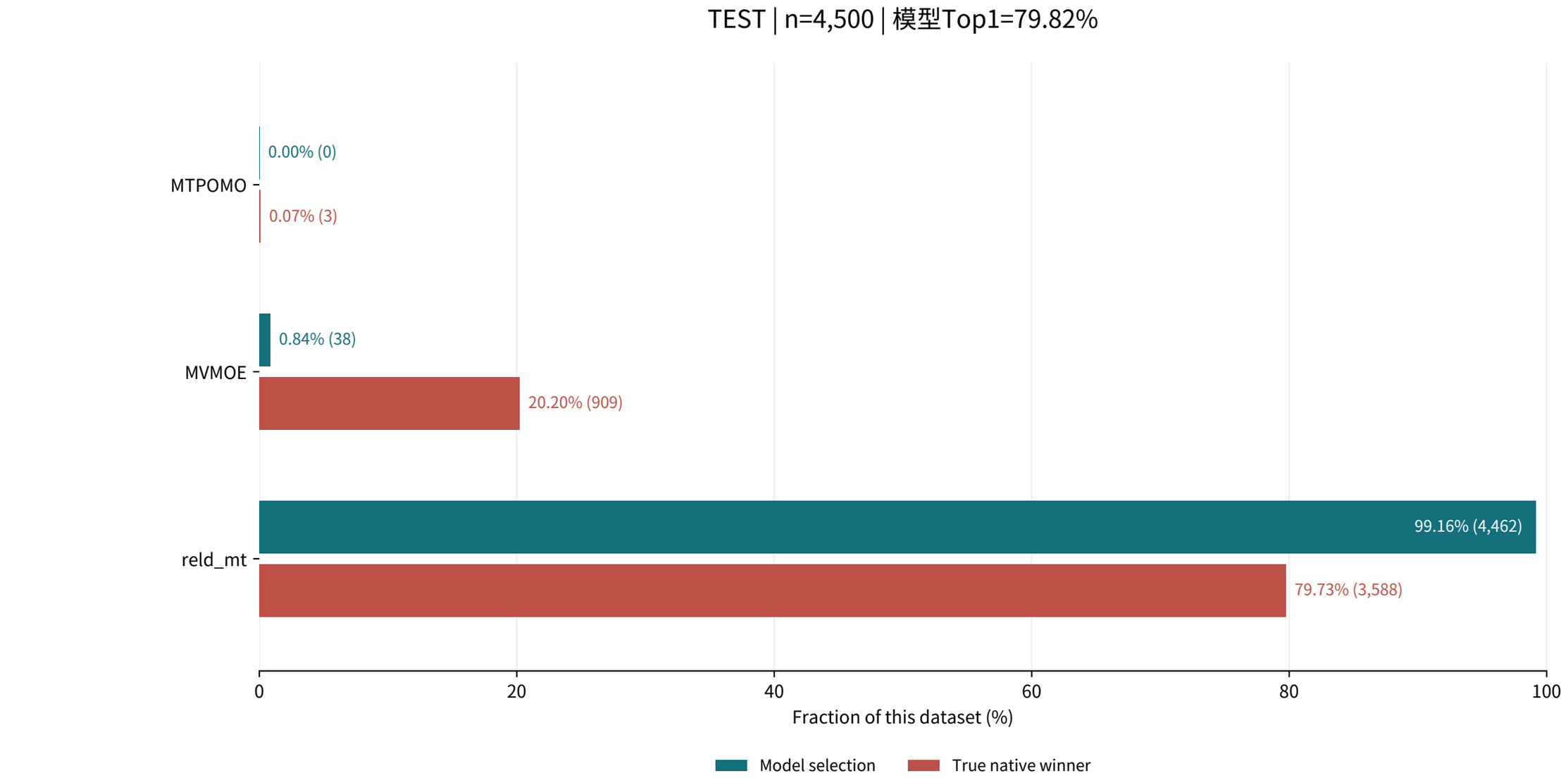
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | VRPBLTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

native int64 ind verified against original FP64 costs; never FP32 argmin



Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

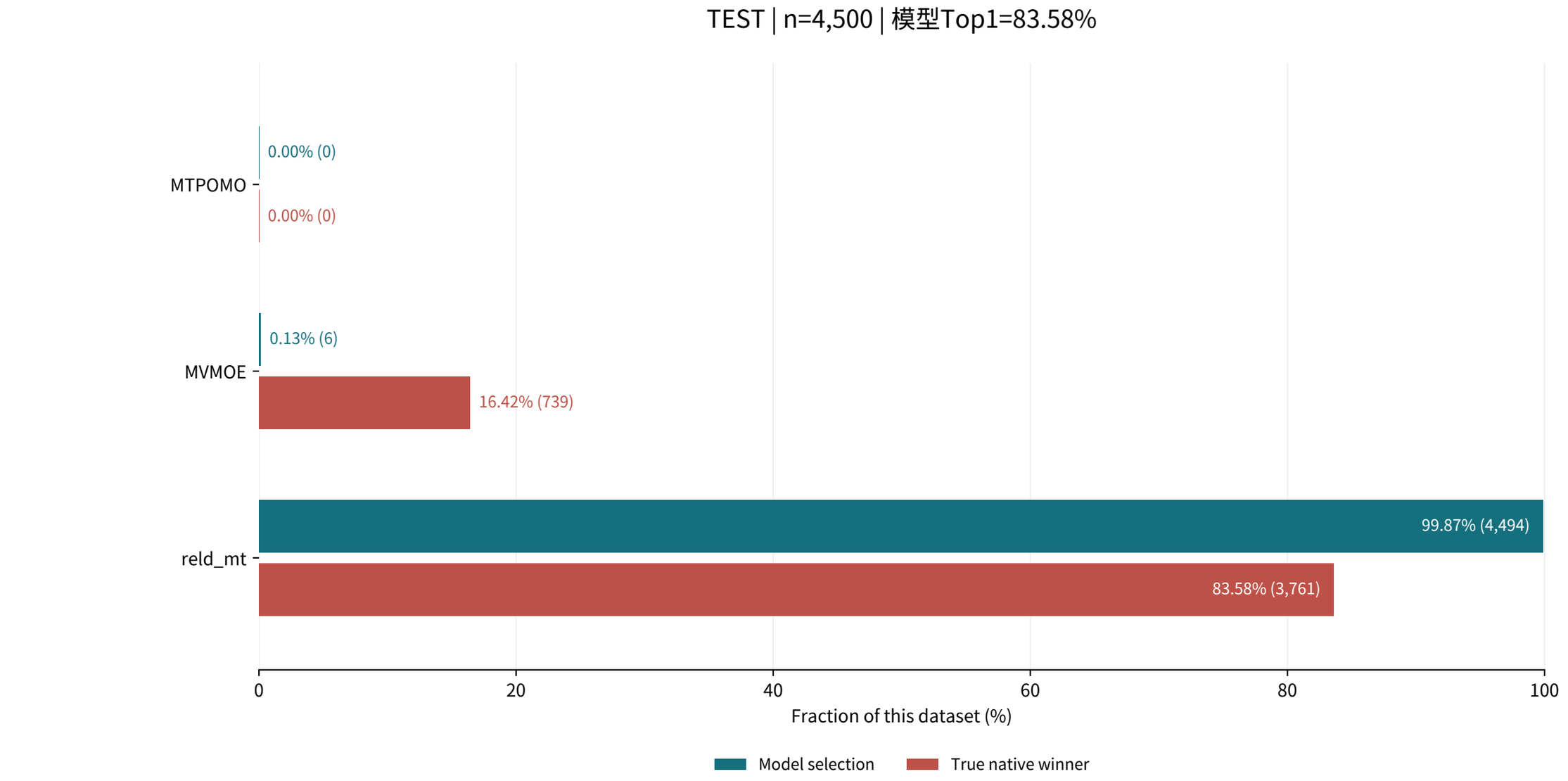
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

R122/model | OVRPBLTW | method choice vs native winner

Same checkpoint f342aedf161db09a0883 | selected update 14400

TEST data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

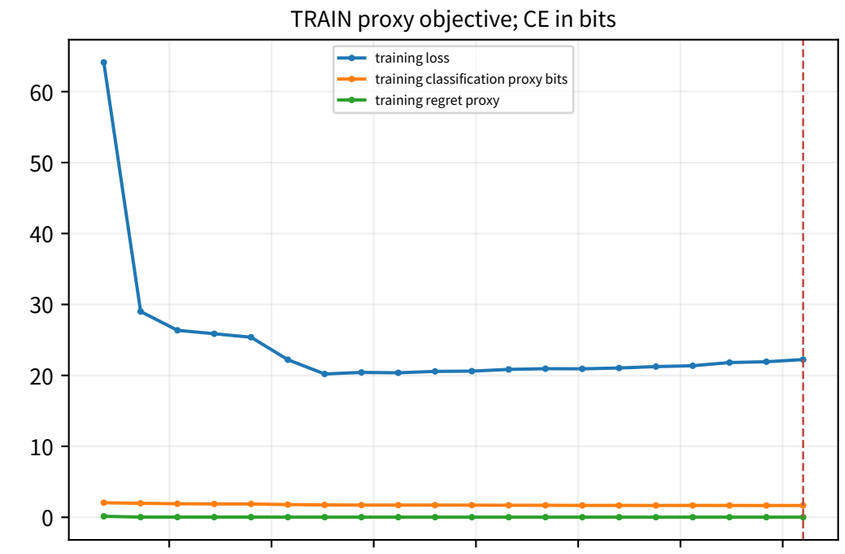
native int64 ind verified against original FP64 costs; never FP32 argmin

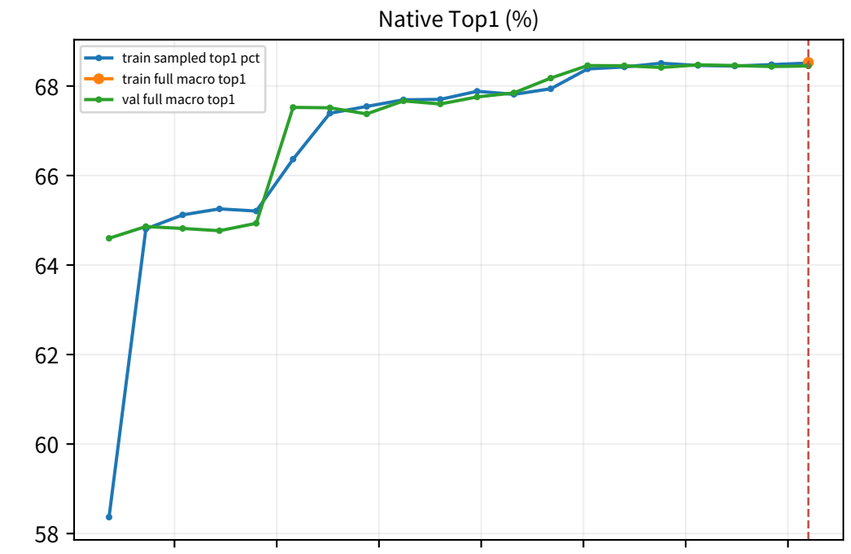


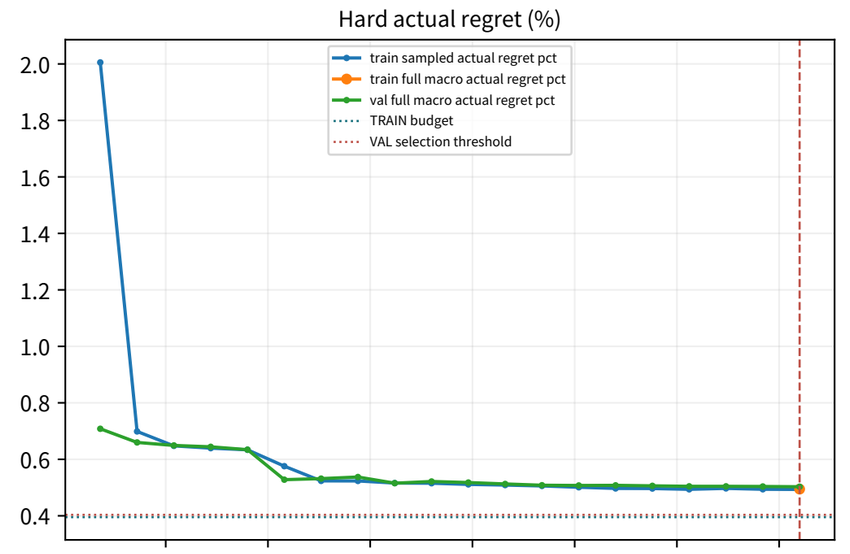
Paired bars share one axis and fixed candidate order. Labels show percent (count), including zeros.

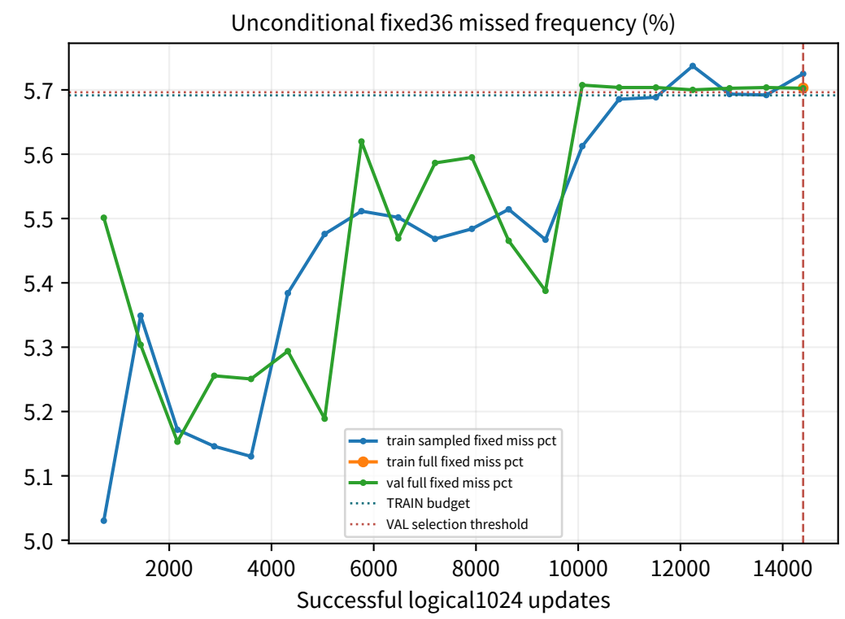
Model Top1 = per-instance native winner agreement, NOT the largest selection frequency. TEST precedes LIB.

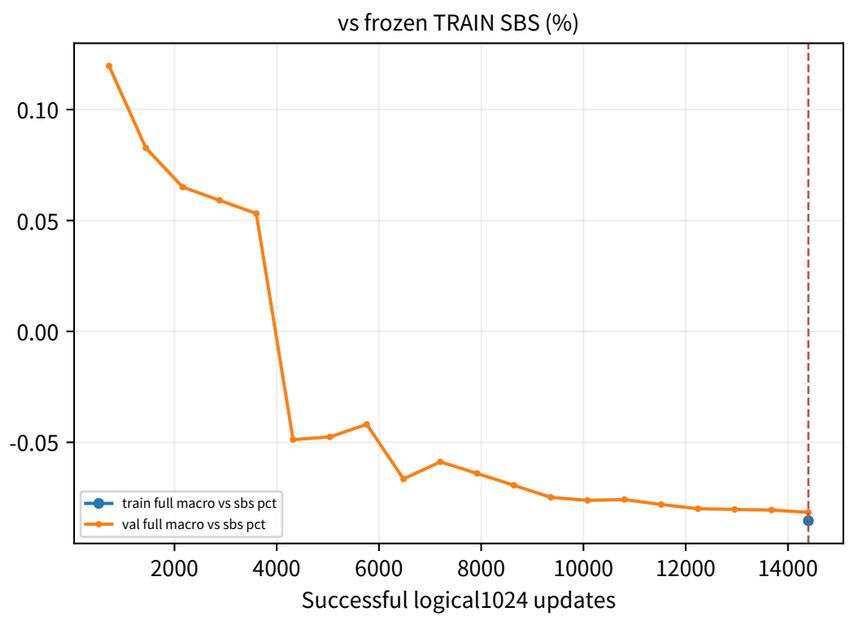
D2: actual TRAIN optimization feedback and full VAL history; one locked BASE

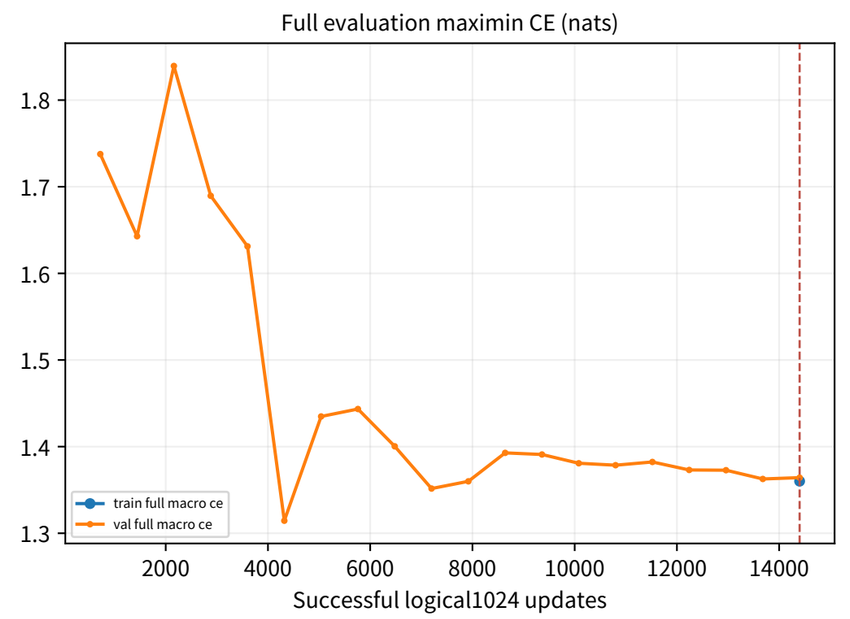












Sampled TRAIN feedback is NOT full TRAIN evaluation. Full TRAIN points exist only where actually saved. No interpolation or TEST/LIB selection.